Mechanical Systems and Signal Processing 251 (2026) 114179

Contents lists available at ScienceDirect

Mechanical Systems and Signal Processing

journal homepage: www.elsevier.com/locate/ymssp

Full Length Article

Neural operators for ultrasonic full waveform inversion: A next-generation ultrasonic NDT with improved fidelity, efficiency, and robustness

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ARTICLE INFO

Keywords:
Fourier neural operator
Fullwaveforminversion
Ultrasonic quantitative detection
Computational wave imaging
Deep learning

ABSTRACT

Full waveform inversion (FWI), a computational wave imaging technique that fully incorporates the physics of wave propagation, enables high-resolution quantitative reconstruction of inhomogeneous media and promotes effective application in non-destructive testing (NDT). However, its widespread adoption is hindered by the intrinsic challenges of solving a nonlinear, ill-posed inverse problem, resulting in a high computational cost considering the iterative implementation, sensitivity to the initial model, and the possible convergence to a local minimum solution. Recent advances in data-driven FWI frameworks using neural networks have improved both accuracy and efficiency, yet challenges in robustness and generalization persist. Neural operators, an emerging paradigm in scientific machine learning, aim to learn mappings between infinite-dimensional function spaces, offering superior generalization and scalability for complex physical systems. Motivated by this, we propose a neural-operator-based framework for direct inversion of ultrasonic full-waveform data based on the Unet Fourier Neural Operator (U-FNO), extending the neural operator paradigm to ultrasonic NDT. By leveraging parameterized Fourier kernels to globally capture intricate wave physics, U-FNO demonstrates improved inference efficiency relative to classical FWI, and improved fidelity and robust reconstruction compared with representative data-driven models and total focusing method (TFM). Numerical and experimental results demonstrate that the proposed framework provides a promising direction for next-generation computational wave imaging technology in ultrasonic NDT.

1. Introduction

Acoustic imaging seeks to infer hidden internal structures and unknown physical properties of materials by analyzing measured wave signals[1]. It plays a critical role in a wide range of applications, including seismic inversion, underwater object detection, biomedical imaging, and non-destructive testing (NDT). This task typically involves solving ill-posed and nonlinear inverse problems. In particular, sensor data are generally available only at the surface or boundaries of an object, leading to a dimensional mismatch between the limited observed data and the unknown parameter space. Additionally, the inherent inhomogeneity of the medium gives rise to complex wave phenomena such as diffractions, scattering, and reflections. As a result, a single set of observations may

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https://doi.org/10.1016/j.ymssp.2026.114179
Received 26 September 2025; Received in revised form 4 March 2026; Accepted 18 March 2026
Available online 7 April 2026
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correspond to multiple physically plausible solutions, thereby increasing both the difficulty and uncertainty of the inversion process.

To address these challenges, various imaging methods have been developed, as shown in Fig. 1. Early approaches, primarily based on ray approximations, are computationally efficient and stable. Representative techniques include the synthesis aperture focusing (SAFT)[2], the total focusing method (TFM)[3], the probabilistic diagnostic imaging (PDI)[4], and plane wave imaging (PWI). However, these methods are inherently limited to qualitative reconstruction, as the resulting images only approximate the spatial variation of acoustic parameters. Ray-based tomography [5], which estimates the sound velocity distribution of a medium based solely on travel time, offers coarse quantitative analysis. It ignores wave diffractions and scattering, restricting resolution to the width of the first Fresnel zone[6,7]. To improve ray-based methods while maintaining computational efficiency, wavefield extrapolation methods such as phase shift migration (PSM) [8] and Stolt migration[9] adopt a one-way wave approximation to partially account for diffraction effects. Diffraction tomography, leveraging linearized scattered field approximations (e.g., Born or Rytov approximations), accounts for both diffraction and scattering[10] and enhances the resolution for simple defects. However, its applicability remains confined to weak scattering and fails for complex defects with high velocity contrasts[11].

Full-waveform inversion (FWI) has emerged as a powerful method for quantitative subsurface reconstruction with high resolution and accuracy by minimizing the mismatch between observed and synthetic data[12,13]. Its high resolution potential is due to the numerical solution of the exact wave equation, which describes more completely the physics of wave propagation in heterogeneous media, and can be realized with a resolution of half the wavelength of the dominant frequency[14,15]. However, directly transferring FWI to ultrasonic NDT is challenging due to differences in frequency range, wave propagation characteristics, and the high acoustic impedance contrast of materials. As a result, although FWI holds great promise for advancing ultrasonic NDT, its application has thus far been restricted to a small number of preliminary studies. For instance, Rao et al.[16] developed an acoustic multiparametric FWI method for the reconstruction of velocities and densities of metallic components. Further, they extended it to quantitative imaging of guided-wave corrosion damage[17]. In addition, He et al.[18] investigated the application of FWI for reconstructing complex metal components in a water bath environment. Their numerical evaluation demonstrated FWI has significant potential for accurately recovering complex structural features. To further validate these findings, experimental studies on homogeneous materials were subsequently conducted using accurate transducer modeling [19]. Bürchner et al.[20] quantitatively compared TFM, RTM, and FWI methods for phased array full matrix capture (FMC) data across multiple metrics, highlighting the powerful potential of FWI for defect identification and sizing. However, the FWI approach is computationally expensive, as each iteration requires forward modeling the full wavefield in a nonlinear optimization framework. Moreover, its inherent nonlinearity and ill-posedness often lead to convergence to local minima, making reconstructions highly sensitive to the quality of the initial model[13]. To address the aforementioned issues, several physics-based approaches have been developed. These approaches encompass the development of regularization methods[21], the utilization of low-frequency information to construct an improved initial model[22], the refinement of the convexity of the objective function[23], and Multi-scale optimization strategies[24]. Despite these advancements, physics-based FWI has not consistently succeeded in ultrasonic NDT inversion, underscoring the need for alternative solutions.

Motivated by the rapid development of deep learning, which has demonstrated a strong ability to establish nonlinear mappings between different data domains, researchers have suggested data-driven direct inversion methods to alleviate the optimization burden of conventional FWI[25,26]. The primary objective of these methods is to transform the optimization problem of FWI into a supervised learning task. This task involves constructing an inverse mapping from observed data to the Measurement structure by training neural networks on paired data of waveforms and medium acoustic parameters, also referred to in the literatures as data-driven FWI[27–30]. Data-driven direct inversion methods based on different neural network architectures have achieved notable success in seismic exploration, particularly in terms of computational efficiency and inversion accuracy[25–31]. More recently, such methods have been explored in ultrasound imaging. Suo et al.[32] developed an improved dual-encoder-based Unet with high-frequency feature enhancement for the quantitative imaging of bone sound velocity and density, which addresses the highly nonlinear challenge of FWI posed by high-contrast bone models. Lozenski et al.[33] further highlighted the computational advantages of learning FWI using CNN, achieving reconstruction speeds three orders of magnitude faster than traditional FWI. Ren et al.[34] proposed a physics-embedded neural network framework for accurate quantitative reconstruction of tumors in brain tissue by training forward convolutional networks and inverse sub-neural networks to replace the finite-difference modeling and iterative optimization process of FWI. Rao et al.[35]proposed a full convolutional network (FCN) tailored to quantitatively detect defects in multi-layered bonded composites using FMC. Liu et al.[36] developed an Unet-based method to fully utilize echo space represented by FMC to achieve super-resolution limit and sub-wavelength defects visualization. Data-driven methods may be superior to traditional physics-based methods when the

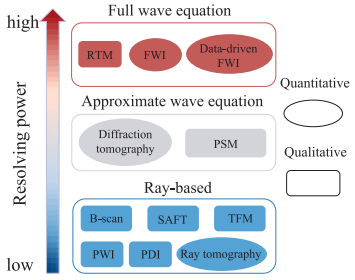


Fig. 1. Acoustic imaging methods based on different theoretical assumptions.

training and prediction data share similar distributions. However, even moderate distribution shifts between training and prediction data may lead to considerable performance degradation[37]. Therefore, although data-driven methods can reduce computational costs and mitigate local minimum problems compared to physics-based methods, their robustness and generalization capabilities remain areas worthy of further research.

Recently, a new paradigm known as neural operators has been developed to directly learn mappings between infinite-dimensional function spaces[38–40]. Neural operators have been proven to have the powerful ability to accurately approximate both nonlinear continuous functions (mapping from a function space to real numbers) and nonlinear operators (mapping from one function space to another)[38], thereby accelerating scientific simulation and engineering design in fields such as weather forecasting[41] and carbon dioxide storage[42]. Among various neural operator architectures, the Fourier neural operator (FNO)[40], which uses the fast Fourier transform as an integral operator to learn the integral kernel in Fourier space from data, has demonstrated superior accuracy and computational efficiency compared to other neural operator frameworks across a range of benchmark problems[43]. Unlike CNNs restricted to local convolutions on discretized grids, FNO leverages global neural integral operators to learn continuous function instead of discretized vectors, as shown in Fig. 2. This property underpins its strong generalization, high accuracy and resolution-invariance[40]. in wave propagation modeling, effectively capturing complex physical phenomena such as nonlinear effects[44], viscoelasticity[45], and quantum mechanics[46]. Despite these advantages in forward modeling, the application of neural operators to inverse problems remains relatively underexplored. Initial efforts have recently begun to investigate their potential in seismic inversion. Lu et al.[47] incorporated FNO as the decoder within DeepONet to perform seismic inversion with variable source parameters, demonstrating strong robustness to Gaussian noise in observed data. More recently, Huang et al. proposed a neural operator-based FWI framework that employs a pretrained FNO as a surrogate forward model [48] and integrates wave-equation constraints into the inversion objective, improving reconstruction quality while maintaining computational efficiency[49].

Inspired by these developments, we further advance the neural operator paradigm to ultrasonic NDT, addressing the highly ill-posed inverse problem of FWI using phased array FMC data. We propose a U-FNO framework that integrates kernel parameterization in the Fourier domain to capture global wave defect interactions while retaining localized feature recovery through a U-Net path. This design enables the accurate mapping from FMC datasets, including scattering, diffraction, and reflection signals, to the longitudinal wave velocity distribution of complex medium, thereby accelerating inversion efficiency. Compared with existing data-driven FWI methods such as InversionNet[30], U-FNO achieves higher fidelity and robustness in reconstructing defects with high contrast and complex geometries.

The remainder of this paper is structured as follows. Section 2 provides background information on physics-based and data-driven FWI, followed by details on the design of the U-FNO. Section 3 presents the results of complex geometric defect inversion imaging and compares the reconstruction accuracy and robustness of U-FNO and InversionNet. Section 4 validates the method with experimental data and compares the reconstruction results with InversionNet and TFM. Section 5 discusses the limitations and directions for future research. Finally, we share our conclusions in Section 6.

2. Methods

We first introduce the physics-based and data-driven frameworks for full waveform inversion, and then we detail the network framework for Unet Fourier neural operators.

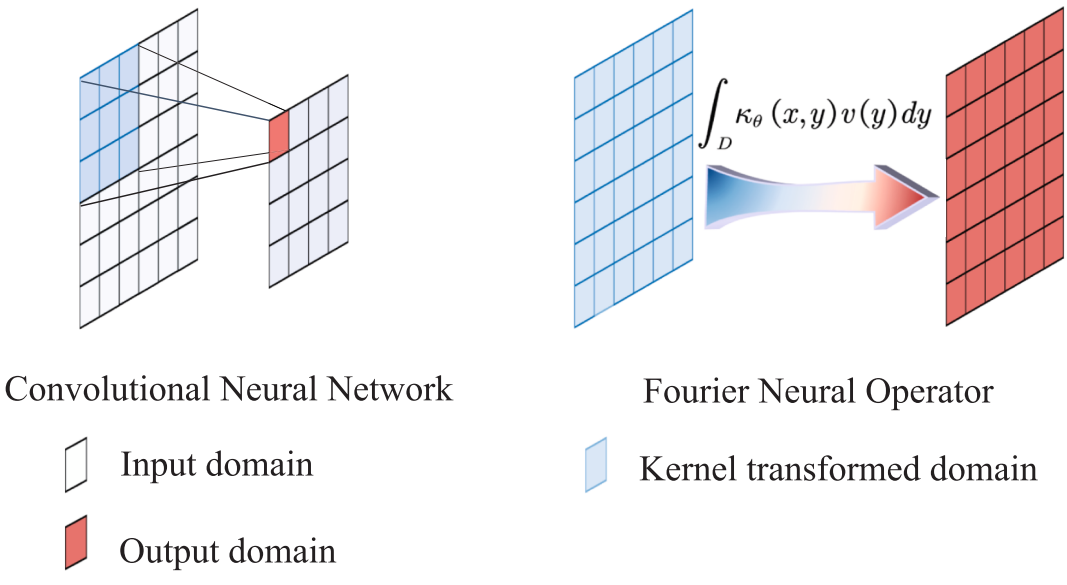


Fig. 2. The difference between Convolutional neural networks (CNNs) and Fourier neural operators.

2.1. Physics-based FWI

FWI is typically based on elastic or acoustic wave equations that consider the effects of wave propagation in inhomogeneous media and establish a relationship between observed waveform data at our sensors and a model of medium parameters (e.g., speed of sound, density, and acoustic impedance) to reconstruct images of parameter distributions.

Waveform forward modelling in 2D provides an accurate description of wave propagation and is the key to waveform inversion algorithms. In the time domain, the two-dimensional constant density isotropic acoustic wave equation is described as follows:

$$\frac{\partial^2 u(x,y,t)}{\partial t^2} - \frac{1}{c(x,y)^2}\left(\frac{\partial^2 u(x,y,t)}{\partial x^2} + \frac{\partial^2 u(x,y,t)}{\partial y^2}\right) = s(x,y,t), \tag{1}$$

where $u(x,y,t)$ represents the acoustic pressure wavefield.$c(x,y)$ stands for the spatial distribution of the sound velocity in an acoustic medium in 2D, and $s(x,y,t)$ is source function. In this paper, we use the finite difference time-domain method (FDTD) to solve the wave equation[50].

The forward modeling problem present in Eq. (1) can be simplified as:

$$u = f(\boldsymbol{m}), \tag{2}$$

where u is acoustic pressure wavefield, f represents the forward modeling operator, and $\boldsymbol{m}$ is the velocity model parameters. The inversion process commences with a specified initial model of the medium parameters, and the synthetic wavefield of the initial model is obtained by forward modelling. The mismatch between the experimentally measured observed wavefield and the synthetic wavefield is employed as an objective function to update the medium model parameters, thereby converging on a solution that generates synthetic data that are consistent with the observed data. The objective function of the inversion process can be described as follows:

$$E(\boldsymbol{m}) = \min_{\boldsymbol{m}}\Big(\|\boldsymbol{d} - f(\boldsymbol{m})\|_2^2 + \lambda R(\boldsymbol{m})\Big), \tag{3}$$

where $\boldsymbol{d}$ is an obverse waveform data set, λ is regularization weight parameter and $R(\boldsymbol{m})$ represents regularization term. $\|\boldsymbol{d} - f(\boldsymbol{m})\|_2^2$ stands for the L_2 norm of data mismatch. The regularization term is used to mitigate the ill-posedness of physics-based FWI and stabilize its convergence. The ill-posedness is primarily attributable to incomplete data coverage and parameter cross-talk. Typically, experimental measurements are conducted on the boundaries of the object to be measured, resulting in limited observed data. Moreover, observed wavefield changes are due to the coupling of several acoustic parameters, making it difficult to accurately resolve the contribution of individual acoustic parameters to the observed wavefield. The update of model parameters can be formulated as:

$$\boldsymbol{m}_{k+1} = \boldsymbol{m}_k - \alpha_k g_k, \tag{4}$$

where $\boldsymbol{m}_k$ represents the velocity model parameters for the k-th iteration, α_k is the step length for the k-th iteration, $g_k = \frac{\partial E(\boldsymbol{m}_k)}{\partial \boldsymbol{m}}$ is the gradient for the k-th iteration. The resolution of the FWI problem typically employs the adjoint-state method to compute the gradient, necessitating forward wave propagation, backward wave propagation, and full-wavefield crosscorrelation in the time domain [13]. This method is computationally intensive and can require a significant amount of time to reach convergence. In this paper, Deepwave is employed, and the gradients are computed through reverse-mode automatic differentiation, which is mathematically equivalent to the classical adjoint-state method[51]. Second-order optimization methods have been shown to accelerate the convergence rate by computing the Hessian matrix of the model parameters of the objective function[52]. However, these methods require additional computational resources.

2.2. Data-driven FWI (Data-driven direct inversion)

Ultrasonic FWI aims to reconstruct the acoustic parameters distribution of the object under test, often for the detection of defects such as holes or through-thickness cracks in metallic materials. These defects can be simplified as regions of sound velocity anomalies (e.g., air inside a hole, where the sound velocity reduces to about 340 m/s). By abstracting the complex defect structure into a non-uniform medium with varying sound velocity, the problem of defect detection is transformed into an inverse reconstruction of a high-contrast sound velocity model. Due to the high nonlinearity and ill-posedness of the objective function and its sensitivity to the initial model, the traditional physics-based FWI method is prone to falling into local minima and struggles to accurately invert high-contrast models, especially when low-frequency information is absent or the initial model is unreliable. In contrast, the data-driven approach addresses these challenges by transforming the problem into an end-to-end mapping task. The key idea of the data-driven approach is to create a training dataset of paired data consisting of velocity model $\boldsymbol{m}$ and corresponding waveform data $\boldsymbol{d}$. This training dataset is then used to train a deep neural network to learn the complex nonlinear mapping from $\boldsymbol{d}$ to $\boldsymbol{m}$, allowing for direct estimation of the velocity distribution from the actual observed data on the measured object. This practice can be formulated mathematically as follows:

$$\begin{aligned} \widehat{\boldsymbol{m}} &= h_{\theta^*}(\boldsymbol{d}) \approx f^{-1}(\boldsymbol{d}), \\ s.t.\theta^*(\Phi) &= \underset{\theta}{\operatorname{argmin}} \sum\nolimits_{(\boldsymbol{m}_i,\boldsymbol{d}_i)\in\Phi} \mathscr{L}(h_\theta(\boldsymbol{d}_i), \boldsymbol{m}_i), \end{aligned} \tag{5}$$

where θ are the trainable parameters for the neural network,$h_{\theta^*}(\cdot)$ is the inversion operator with parameters θ^*, $\mathscr{L}(\cdot,\cdot)$ represents a loss function, and Φ is the training dataset with paired samples of $(\boldsymbol{m_i}, \boldsymbol{d_i})$. Compared to traditional FWI methods, this method is characterized by generating a set of inverse operators rather than a single model inversion. This means that once training is complete, when changing the reconstruction object, the neural network can quickly infer the answer due to the self-efficient large-scale parallel computing architecture, without the need for re-iterative optimization.

2.3. U-FNO architecture

The goal of Fourier neural operator is to learn a mapping between two infinite dimensional spaces from a finite set of observed input–output pairs. In the context of ultrasonic FWI, U-FNO is used to learn an inverse operator mapping of Eq. (5) from paired datasets of velocity model $\boldsymbol{m}$ and FMC $\boldsymbol{d}$. For Eq. (5), we use for the loss function L1 norm.

We discretize the input function in a finite-dimensional space, and specifically evaluate each predetermined exciter positions $D_E = \{x_1, x_2, x_3 \cdots x_E\} \subset D$, receiver positions $D_R = \{x_1, x_2, x_3 \cdots x_R\} \subset D$ at the surface and each receiver records at T time samples $D_T = \{t_1, t_2, t_3 \cdots t_T\} \subset D$. Therefore, the discrete ultrasonic data for each exciter is a matrix of shapes (T, R), The total observed data in discrete form is a tensor of shape (T, R, E). The output m of U-FNO is a function in the spatial domain, so it also needs to be discretized in a finite-dimensional space. We evaluate $\boldsymbol{m}$ at the position W $\times$ H, where W and H correspond to the horizontal and vertical ranges of the velocity map, respectively.

U-FNO is formulated as an iterative architecture as

$$\mathscr{G}_\theta : \boldsymbol{d} \mapsto z_0 \mapsto z_1 \mapsto \cdots \mapsto z_{N-1} \mapsto \boldsymbol{m}, \tag{6}$$

where z_j for $j = 0, 1, \cdots, N-1$ is a sequence of functions. Considering the balance of computational efficiency and performance, we use one Fourier layer (Fig. 3(b)) and two U-Fourier layers (Fig. 3(c)) in U-FNO (Fig. 3(a)).

First, the input observation data d is elevated to a higher-dimensional space through a linear transformation P.

$$z_0 = P(\boldsymbol{d}) \in \mathbb{R}^{T \times R \times C}, \tag{7}$$

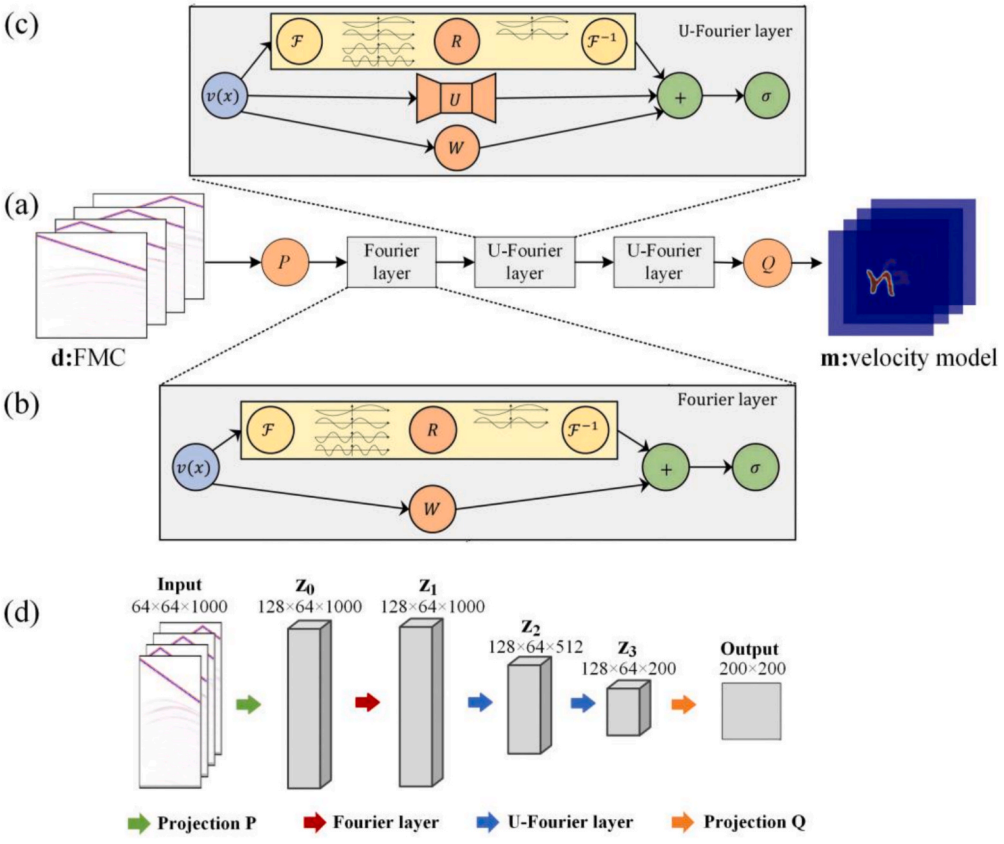


Fig. 3. (a) The U-FNO model architecture. $\boldsymbol{d}$ is the input, P and Q are fully connected neural networks, and $\boldsymbol{m}$ is the output. (b) Fourier layer, adapted from[40]. (c) U-Fourier layer, adapted from [53]. (d) Shapes of outputs..$z_j, j = 0, 1, 2, 3$

where C is the number of hidden channels, controlling the dimensionality of the latent feature space in which the operator is learned, z_0 is the output after passing through the linear layer. We map a function of dimensions T × R × C to W × H × 1 to generate the velocity map $\boldsymbol{m}$. However, the input and output dimensions are the same for the Fourier and Unet layers. To achieve this conversion, we add a linear layer before the activation layer of each Fourier layer and U-Fourier layer.

Next, we use one Fourier layer and two U-Fourier layers as a merged network:

$$\begin{aligned} z_1 &= \sigma\big(W_0'(\mathscr{K}_0(a;\theta)z_0 + W_0 z_0)\big), \\ z_2 &= \sigma\big(W_1'(\mathscr{K}_1(a;\theta)z_1 + \mathscr{U}_1 z_1 + W_1 z_1)\big), \\ z_3 &= \sigma\big(W_2'(\mathscr{K}_2((a;\theta)z_2 + \mathscr{U}_2 z_2 + W_2 z_2))\big), \end{aligned} \tag{8}$$

where z_1, z_2 and z_3 correspond to the outputs of Fourier, U-Fourier 1, and U-Fourier 2 in Table 1. W_i and W_i' are linear transformations. $\mathscr{U}_i$ is a Unet CNN operator. σ is a non-linear activation function.$\mathscr{K}_i(a;\theta)$ is a kernel integral transformation parameterized by a neural network. This is specifically described in neural operators as follows:

$$\big(\mathscr{K}(a,\theta)z_j\big)(x) = \int_D \kappa_\theta(x-y,a)z_j(y)dy, \forall x \in D. \tag{9}$$

The main feature of the FNO method is that it formulates the operator learning task by directly parameterizing the integral kernel in Fourier space. The integral kernel is represented by the fast Fourier transform(FFT) as

$$\big(\mathscr{K}(a,\theta)z_j\big)(x) = \mathscr{F}^{-1}\big(\mathscr{F}(\kappa_\theta)\cdot\mathscr{F}\big(z_j\big)\big)(x) = \mathscr{F}^{-1}\big(R_\theta \cdot \mathscr{F}z_j\big)(x), \tag{10}$$

where $\mathscr{F}$ represents the a two-dimensional FFT,$\mathscr{F}^{-1}$ is its inverse.R_θ is the Fourier transform of a periodic function $\kappa_\theta : \overline{D}\rightarrow\mathbb{R}^{d_z\times d_z}$ parameterized by θ. Notice κ_θ is a periodic function providing Fourier series expansion and works in the discrete mode $k \in \mathbb{Z}^d$ of the Fourier transform. First, we truncate the at the maximum number of modes $k_{\max} = |Z_{k\max}| = \big|\{k \in \mathbb{Z}^d : |k_n| \le k_{\max,j}, \text{for} n = 1,\cdots,d\}\big|$, and then directly parameterize R as a complex-valued tensor of shape $(k_{\max}\times d_z \times d_z)$ with truncated Fourier coefficients therefore drop θ from our notation. Multiplication by the weight tensor R is

$$\big(R\cdot(\mathscr{F}z_j)\big)_{k,l} = \sum_{i=1}^{d_v} R_{k,l,n}(\mathscr{F}v_t)_{k,n}, k = 1,\ldots,k_{\max}, n = 1,\ldots,d_v. \tag{11}$$

In practice, we have found that choosing $k_{\max} = 40$ to be sufficient for all the tasks that we consider. For more details about FNO, please refer to[40,53]. This signifies that the network parameters are defined and learned in Fourier space, and the coefficients corresponding to the Fourier series representation of the output function are inferred from the dataset. This discrete invariance property also enables zero-shot super-resolution techniques[40].

In contrast to the conventional Fourier layer, the U-Fourier layer incorporates the Unet path. The Fourier layer, due to the truncation of Fourier basis functions to a finite number, endows the network with noise resistance; however, this may result in the loss of high-frequency information in the data. However, the Unet path has the capacity to enhance the representational capabilities of the network architecture at higher frequencies by processing data on structured grids through local convolutions[53].

Finally, applying the nonlinear transformation Q to map the output z_3 of the final U-Fourier layer to the velocity model.

$$\boldsymbol{m} = Q(z_3) \in \mathbb{R}^{W\times H\times 1}. \tag{12}$$

More details of the operations and shapes of U-FNO are summarized in Table 1, and the detailed process is described in Algorithm 1.

Algorithm 1 U-FNO for FWI

Input: $\boldsymbol{d}$: FMC datasets, $\boldsymbol{m}$: true velocity models, $\widehat{\boldsymbol{m}}$: predicted velocity models, N: number of training sets, E: number of epochs, I: number of iterations of each epoch updating velocity models,B: number of batches, lr: learning rate for training

Step1: Consider the U-FNO framework with parameters θ for approximating the inverse operator.

Step2: Generate 2D velocity models from training dataset

Step3: Synthesize FMC data solving Eq.(3) by finite difference scheme

(*continued on next page*)

Table 1
U-FNO architecture. The merged network includes one Fourier layer and two U-Fourier layers.

	Operations	Output shape
Projection P	Linear	(128,64,1000)
Fourier layer	Add (Fourier2d + Conv1d), ReLu	(128,64,1000)
U-Fourier layer 1	Add (Fourier2d + Conv1d + UNet2d), Linear, ReLu	(128,64,512)
U-Fourier layer 2	Add (Fourier2d + Conv1d + UNet2d), Linear, ReLu	(128,64,200)
Projection Q	Permute, Linear, ReLu, Linear, ReLu,	(64,200)
	Permute, Linear, ReLu, Linear	(200,200)

(*continued*)

Algorithm 1 U-FNO for FWI

Step4: Input all data pairs into the network and use the Adam algorithm to update the hyperparameters
for e = 1, …, E **do**
for b = 1, …, B **do**
for n = 1, …, I **do**
Sample training pairs $(\boldsymbol{m}_i, \boldsymbol{d}_i)$ from Φ
Input $\boldsymbol{d}_i$ into the network to obtain $\widehat{\boldsymbol{m}}_i$ given Eq. (6)
Compute loss given Eq. (5).
Update parameters θ using the Adam optimizer with lr: $\theta_{m+1} = \theta_m - lr\nabla_\theta \mathscr{L}(\theta_m)$
end for
end for
end for

2.4. Quality evaluation of imaging results

In this paper, the structural similarity index measure (SSIM), mean absolute error (MAE), root mean square error (RMSE), and L^2 relative error are used to quantitatively evaluate the quality of inverse imaging. SSIM measures the differences between two images based on brightness, contrast, and structure to better reflect the intuitive perception of the human eye [54]:

$$\mathrm{SSIM}(\mathrm{x},\mathrm{y}) = \frac{\left(2\mu_x\mu_y + C_1\right)\left(2\sigma_{xy} + C_2\right)}{\left(\mu_x^2 + \mu_y^2 + C_1\right)\left(\sigma_x^2 + \sigma_y^2 + C_2\right)}, \tag{14}$$

$$\mathrm{MAE}(x,y) = \frac{1}{n}\sum |y - x|, \tag{15}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum (y - x)^2}, \tag{16}$$

$$L_2 = \frac{1}{n}\sum \left|\frac{y - x}{y}\right|, \tag{17}$$

where n is the number of samples. μ_x, μ_y, σ_x,σ_y and σ_{xy} are the local mean, standard deviation, and cross-covariance of images x and y. The regularization constants $C_1 = 10^{-4}$ and $C_2 = 9 \times 10^{-4}$ avoid numerical instabilities when local mean and standard deviation are close to zero. Therefore, greater SSIM, smaller MAE, RMSE and L_2 relative error indicate a better velocity reconstruction.

3. Numerical experiments

3.1. Simulation setup

The Extended Modified National Institute of Standards and Technology (EMNIST) dataset comprises handwritten character digits derived from the NIST Special Database 19 and converted into a 28 × 28 pixel image format[55]. It serves as a core benchmark in machine learning, particularly due to its rich structural variability and visual complexity. Owing to these characteristics, the EMNIST dataset has also found widespread application in computational imaging tasks such as electromagnetic backscattering[56] and ultrasound imaging[57], where it is commonly used to evaluate model performance in handling complex scatterers or targets. In this study, EMNIST patterns are employed to systematically generate geometrically diverse and topologically complex velocity distributions. Such a benchmark, characterized by substantial structural variability beyond simple parametric inclusions, enables rigorous evaluation of the inversion framework for reconstructing complex scatterers and provides a comprehensive assessment of its quantitative inversion capability, robustness, and generalization performance.

We constructed a sound velocity model dataset based on the letters subset of the EMNIST dataset, which includes only handwritten samples of 26 uppercase and lowercase letters, excluding numeric digits. To satisfy the spatial resolution requirements of forward modeling, each 28 × 28 pixel image was zero-padded and expanded to a resolution of 200 × 200 pixels. These high-resolution binary images were then mapped to sound velocity distributions, where pixel intensity values were linearly translated to sound velocities ranging from 1500 m/s to 5800 m/s. This mapping allows each pixel to represent local variations in acoustic properties, effectively simulating complex internal defect patterns typically encountered in industrial inspection scenarios. A total of 6000 sound velocity models were prepared. Among them, 4,500 were utilized for the training of the neural network, 500 were allocated as a validation set, and the remaining 1,000 samples were designated for testing purposes. The EMNIST data format guarantees that each sample is distinct and independent.

To generate the paired dataset (m, d) required for network training, forward wavefield simulations were conducted by solving the acoustic wave equation using a staggered-grid finite difference scheme, based on Eq. (1). The computational domain was discretized with second-order temporal and eighth-order spatial accuracy, using a spatial grid step of 0.2 mm to ensure numerical stability and

precision[50]. The entire computational domain is 40 × 40 mm. Additionally, perfectly matched layers (PML) were implemented at the lateral boundaries to eliminate artificial reflections. As illustrated in Fig. 4, a 64-element linear phased array with an inter-element spacing of 0.6 mm was placed along the model surface. Each array element was sequentially activated to emit a Ricker wavelet with a center frequency of 2.25 MHz, while all elements simultaneously recorded the resulting wavefield. For each transmission, 1000 time samples were collected, resulting in a final data cube of dimensions 64 (transmitters) × 64 (receivers) × 1000 (time steps). This full-matrix capture (FMC) dataset contains rich information about the scattered wavefield induced by complex internal structures. The time step was set to $2 \times 10 - 8$ s, and the total simulation time was $2 \times 10 - 5$ s. It is important to note that only longitudinal waves were excited and recorded, and potential mode conversions between longitudinal and transverse waves were not considered in this study.

3.2. Inversion results

To evaluate the proposed U-FNO network, we used InversionNet as the baseline model. InversionNet employs a fully convolutional network (FCN) consisting of an encoder and decoder to perform the inversion process, and has been proven to be an effective seismic inversion method, achieving state-of-the-art results on several benchmark datasets. The two methods were coded by the PyTorch 1.13 framework in the Python 3.9 environment, and trained using an Adam optimizer under the same parameter configuration as shown in Table 2. The observed data and velocity model were normalized to a range between − 1 and 1 during training and testing. All experiments were completed on a computer equipped with an NVIDIA RTX A6000 graphics card (48 GB) and an Intel 4210R central processing unit (CPU).

Fig. 5 (a), (d), and (g) show three real velocity models randomly selected from the test set. Fig. 5 (b), (e), and(h) are the inversion results of U-FNO, indicating that U-FNO can accurately reconstruct the geometry of different defects. Fig. 5 (c), (f), and(i) depict the inversion results of InversionNet, which can roughly reconstruct the geometry of defects but produces redundant abnormal regions in the details. This indicates that InversionNet may reconstruct inaccurately on new samples that it has not seen before. For a clearer comparison, the vertical velocity profiles extracted from the models in Fig. 5 (a-i) at a horizontal position of x = 21 mm were compared and shown in Fig. 5 (j-m). Fig. 5 (j-m) describes the ground truth (black), U-FNO (blue), and InversionNet (red) predicted vertical velocity profiles.

In almost all cases, the predicted profile values by the proposed method are closer to the true velocity values and more consistent in terms of spatial location. Furthermore, Fig. 6 illustrates the distribution of reconstruction errors for different methods on the test set, indicating that U-FNO demonstrates superior proficiency in capturing intricate spatio-temporal relationships inherent in observed data, thereby exhibiting enhanced accuracy in comparison to InversionNet. In addition, quantitative metrics for evaluating different methods on the test set, including SSIM, MAE, RMSE, and L^2 relative error are listed in Table 3. The numerical results show that the prediction results of U-FNO have smaller MAE, RMSE, L^2 relative and greater SSIM, indicating that this method has higher accuracy and generalization than the baseline method.

To provide a fair comparison with conventional FWI, inversion experiments were conducted using the Deepwave[51] under identical numerical settings and on the same GPU platform as the proposed method, ensuring consistency in both computational configuration and hardware environment. Fig. 7(a)–(c) present the ground-truth velocity models, corresponding to the same

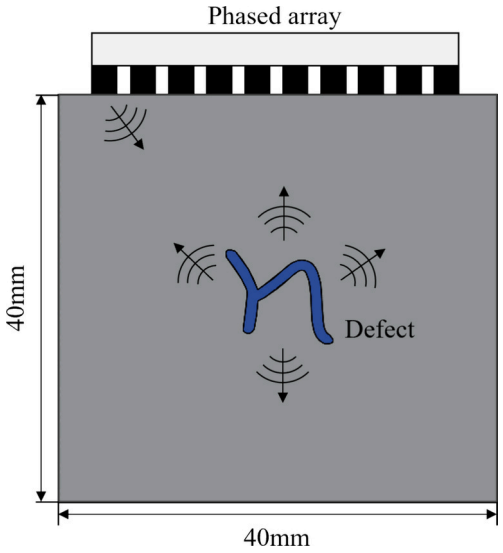


Fig. 4. A schematic plot depicting a velocity model, in which the brown blocks indicate the position of sources and receivers, and blue structure indicates a complex geometric defect model generated from the EMNIST letter dataset. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 2
Network training configuration.

Model	Batchsize	Learning rate	samples	Iterations	k_{max} Fourier modes (truncation order)
U-FNO	32	$1e^{-3}$	4500	60,000	40
InversionNet	32	$1e^{-3}$	4500	60,000	

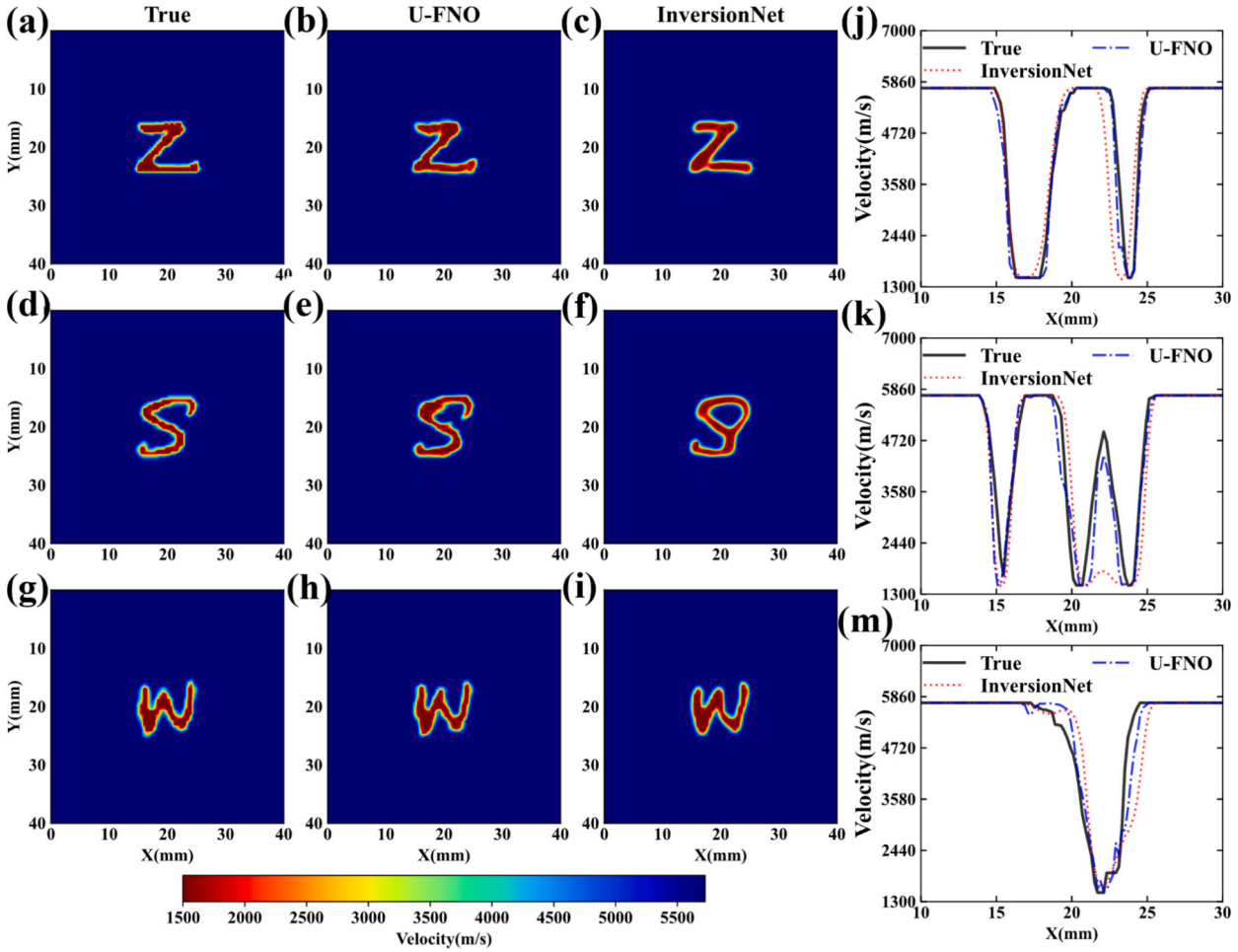


Fig. 5. U-FNO and InversionNet inversion results of the velocity model with complex geometric defects. (a), (d), and (g) show the true models for the inversions. (b), (e), and (h) show the U-FNO inversion results corresponding to (a), (d), and (g), respectively. (c), (f), and (i) show the InversionNet inversion results corresponding to (a), (d), and (g), respectively. (j), (k), and (m) show the one-dimensional velocity profiles at x = 21 mm locations extracted from the true model, U-FNO inversion result, and InversionNet inversion result.

parameter configurations as Fig. 6(a), (d), and (g). Fig. 7(d)–(f) show the FWI reconstruction results obtained from a homogeneous initial model. For high-contrast velocity distributions, the FWI reconstruction does not converge to the true velocity model when initialized from a homogeneous model. In contrast, the proposed framework achieves stable and accurate recovery under the same experimental conditions. Table 4 summarizes the training and inference time comparison among FWI, InversionNet, and U-FNO. The results show that neural-operator-based inversion requires higher computational cost during the offline training phase compared with conventional FWI. Nevertheless, once trained, the learned inverse operator enables substantially faster reconstruction for datasets acquired under similar measurement configurations.

3.3. *Effect of noise on inversion*

In consideration of the pervasive presence of noise in real-world scenarios, the present study investigates the impact of noise on the proposed method. Gaussian noise with SNR = 6 dB and 20 dB is added to the observed data of the velocity model in Fig. 5(g) as two test

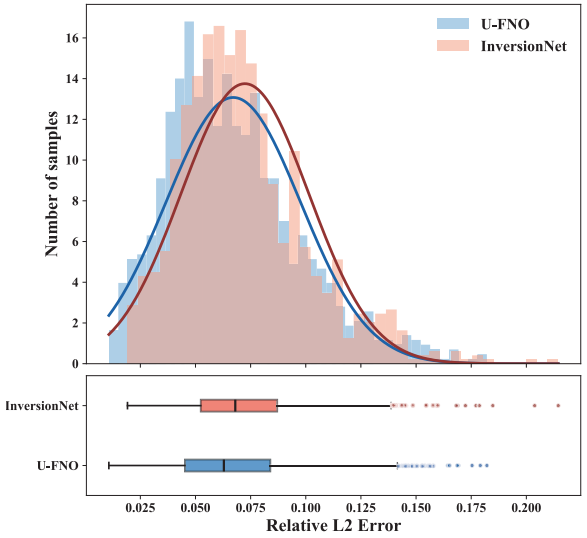


Fig. 6. The distribution of reconstruction errors for 1000 test samples.

Table 3
Quantitative evaluation indicators for inversion results of different models under 1,000 test samples.

Model	MAE	RMSE	SSIM	L^2 relative error	Test time
U-FNO	0.008466	0.0666	0.989006	0.06824	49 s
InversionNet	0.008751	0.07078	0.987948	0.07379	26 s

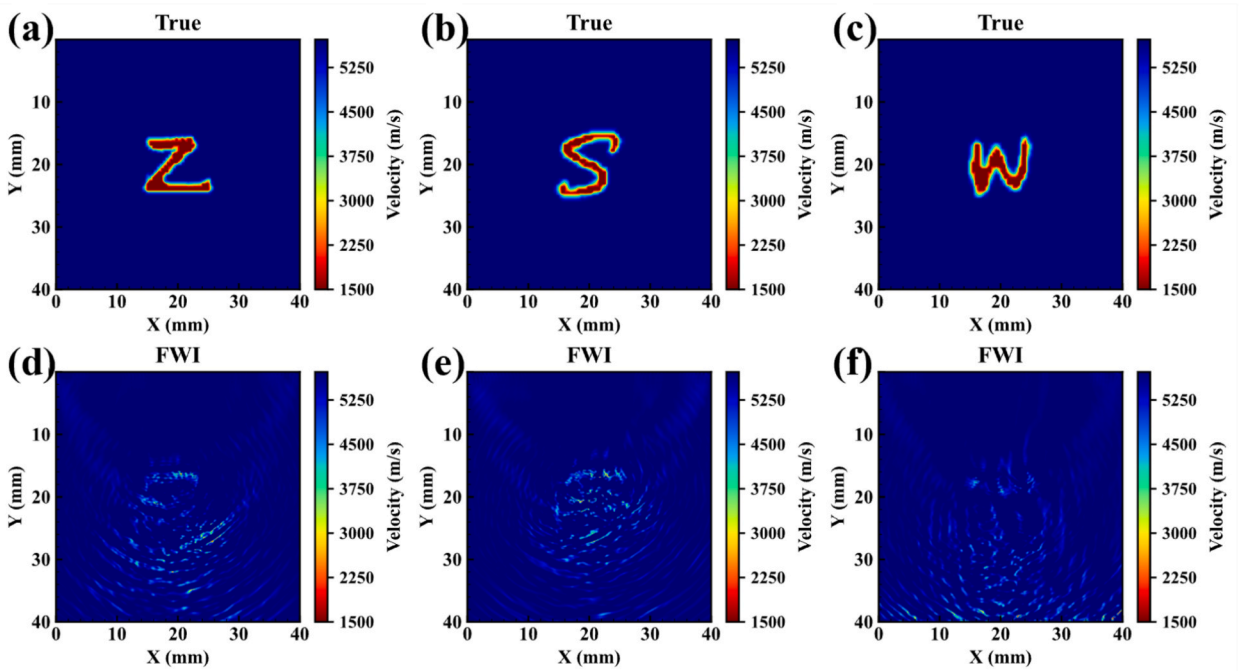


Fig. 7. FWI inversion results of the velocity model with complex geometric defects. (a), (b), and (c) show the true models for the inversions. (d), (e), and (f) show the FWI inversion results corresponding to (a), (b), and (c), respectively.

Table 4
Training and inference time comparison among FWI, InversionNet, and U-FNO.

Model	Iterations	Training time /iteration	inference time per reconstruction
U-FNO	60,000	1.809 s	0.049 s
InversionNet	60,000	1.059 s	0.026 s
FWI	100	72.75 s	7275 s

samples, and these are then input into the network. It is important to note that no retraining is necessary when adding noise to the observed data.Fig. 8 shows the inversion results of U-FNO and InversionNet for two test samples. The true velocity model is shown in Fig. 8(a) and (d), while Fig. 8(b) and (c) show the inversion results of U-FNO for observed data SNR of 20 dB and 6 dB, respectively. This indicates that the proposed method maintains acceptable reconstruction quality under moderate noise levels. Fig. 8(e) and (f) show the inversion results of InversionNet for observed data SNR of 20 dB and 6 dB, respectively, demonstrating that when noise levels are minimal, the network can approximately reconstruct the geometry of the defect; however, as noise levels reach a certain threshold, the reconstruction becomes significantly inaccurate. Similarly, a comparison of the vertical velocity profiles of the reconstructed image at y = 21 mm is shown in Fig. 8(g). Table 5 lists the quantitative metrics of the inversion results for a single model under different SNR levels, including SSIM, MAE, RMSE, and L^2 relative error. The distribution of reconstruction errors for different methods on the test set with 6 dB SNR is shown in Fig. 9. These findings suggest that the proposed method exhibits enhanced robustness and stability.

4. Experimental validation

4.1. Experimental setup

The proposed method was then experimentally verified and compared with the baseline model and the experimental setup is approximately consistent with the simulation. As shown in Fig. 10, the experimental setup uses the Pioneer Phased Array Acquisition System (The Phased Array Company), which is equipped with 64 fully independent channels, enabling high-fidelity signal acquisition and precise control of each array element. The device is connected to a linear phased array probe (2.25 MHz-128elts, IMASONIC, France) operating at a center frequency of 2.25 MHz, with a 0.6 mm element pitch. Three Hastelloy C-276 specimens measuring 150 × 40 × 30 mm (longitudinal wave velocity 5724 m/s) were used. As a typical solid solution strengthened nickel-based high-temperature alloy, Hastelloy C-276 is widely used in nuclear power and petrochemical fields due to its excellent combination of oxidation, corrosion, and high temperature resistance[58]. The machinery of complex geometric defects is achieved through the utilization of wire cutting technology. The configuration of the defects can be referenced in Fig. 11 (a), (d), and(g), which are freely designed using spline curves to ensure that they are different from the training set. The three specimens were immersed in a water bath to establish

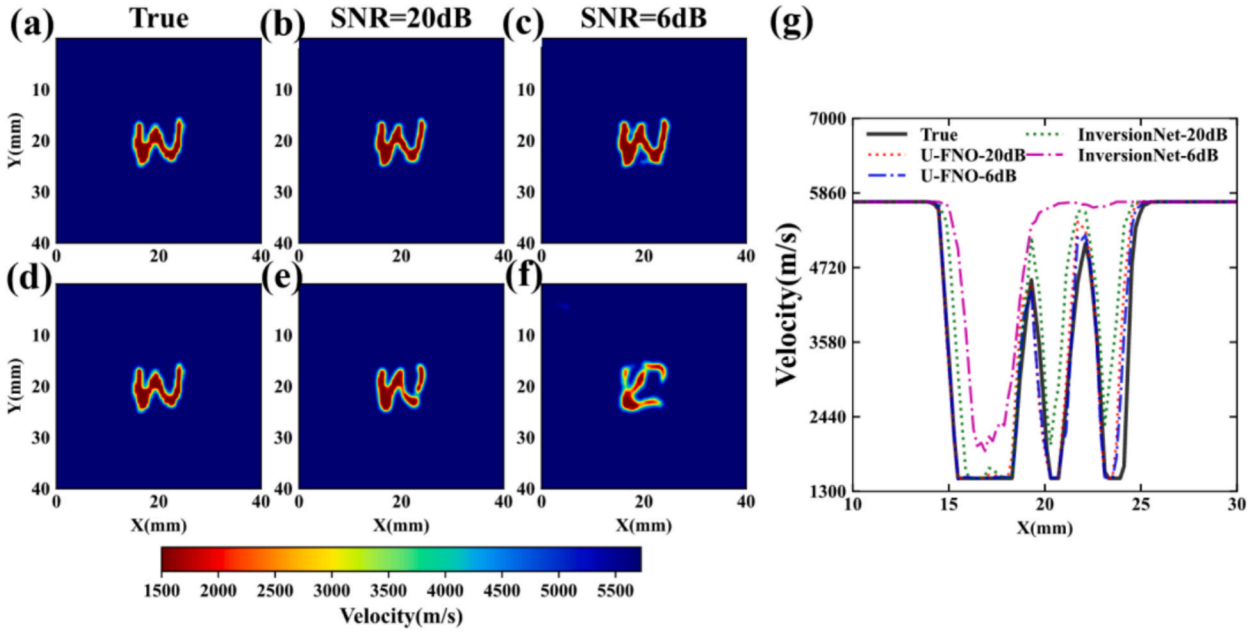


Fig. 8. U-FNO and InversionNet inversion results of the velocity model in Figs. 5 (g). (a) and (d) show the true model for the inversion. (b) and (c) show the U-FNO inversion results of signals with SNR of 20 dB and 6 dB, respectively. (e) and (f) show the InversionNet inversion results of signals with SNR of 6 dB and 25 dB, respectively. (g) shows the one-dimensional velocity profiles at y = 21 mm location extracted from the true model, U-FNO, and InversionNet inversion result.

Table 5
Quantitative metrics of inversion results for a single sample with different SNR in observed data.

Model	SNR	MAE	RMSE	SSIM	L^2 relative error
U-FNO	20 dB	0.007482	0.047775	0.991512	0.054616
	6 dB	0.009676	0.057599	0.987443	0.065848
InversionNet	20 dB	0.010650	0.073255	0.985376	0.083746
	6 dB	0.032916	0.177602	0.955610	0.203037

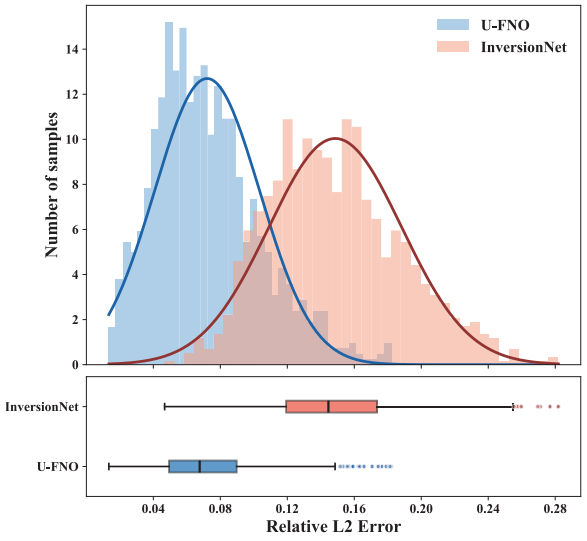


Fig. 9. The distribution of reconstruction errors for 1000 test samples with 6db SNR.

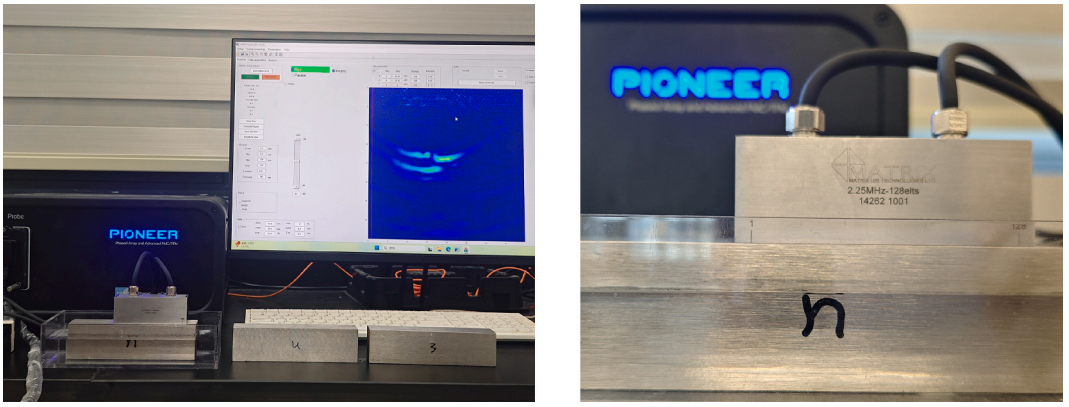


Fig. 10. Experiment setup consisting of a Pioneer Phased Array Acquisition System, a linear phased array, and three specimens.

acoustic coupling between the interfaces, thereby ensuring that sound waves undergo multiple scattering within the defects to provide enhanced defect information, as shown in Fig. 10. A 2.25 MHz Ricker wavelet was used as the excitation signal, and the array data were recorded using FMC. Note that only channels 1–64 were used.

4.2. *Inversion results*

The FMC data of the three test specimens collected in the experiment were used as input for the network, and the inversion results are shown in Fig. 11. Fig. 11 (a) and (d) show two real velocity models through parametric experimental specimens with characteristics similar to the training samples. Fig. 11 (b) and (e) are the inversion results of U-FNO. Fig. 11 (c) and (f) depict the inversion results of InversionNet. Furthermore, a comparison of the horizontal velocity profiles of the reconstructed images at y = 21 mm is shown in Fig. 11 (j) and (k). The findings demonstrate that the experimental and simulated results of the proposed method are in close

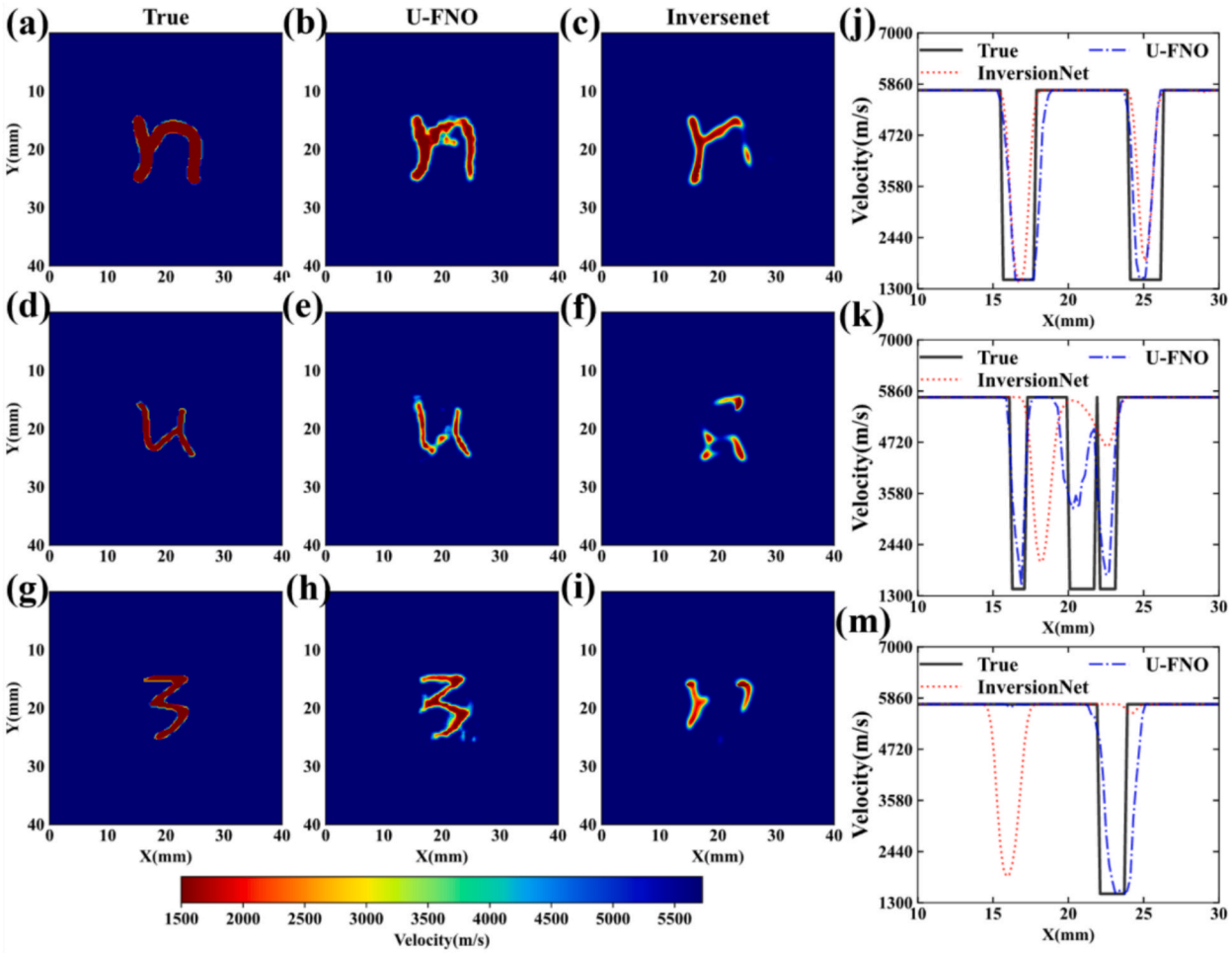


Fig. 11. U-FNO and InversionNet inversion results of the parametric experimental specimens with complex geometric defects in the laboratory. (a), (d), and (g) show the true models for the inversions. (b), (e), and (h) show the U-FNO inversion results corresponding to (a), (d), and (g), respectively. (c), (f), and (i) show the InversionNet inversion results corresponding to (a), (d), and (g), respectively. (j), (k), and (m) show the one-dimensional velocity profiles at y = 21 mm locations extracted from the true model, U-FNO inversion result, and InversionNet inversion result.

agreement, achieving high-precision reconstruction of high-contrast sound velocity models. In contrast, InversionNet is only capable of reconstructing a portion of the defect geometry. Since these network models share the same training samples and training strategies, accurate inversion results indicate that the proposed Fourier layer and U-Fourier layer make significant contributions to the accurate reconstruction of defect velocity models. Specifically, the regularization effect introduced by the Fourier finite group truncation can enhance the model's robustness, stability, and generalization performance. Furthermore, the Unet path has the capacity to facilitate more accurate restoration of defect details by capturing lost high-frequency features.

Although the proposed method accurately reconstructs the overall defect geometry, residual quantitative errors and localized artifacts are still observed in the predicted velocity models. A possible reason is that the proposed method is based on the acoustic wave equation, which means that shear waves and modal transitions contained in the experimental data are not considered during training, resulting in this information being misinterpreted as defects and sound velocity anomalies. Note that although the elastic wave equation is more suitable than the acoustic wave equation for describing material behavior, the acoustic equation can produce a good kinematic L-wave approximation of the elastic wave equation, which not only reduces computational costs but also facilitates the inverse algorithm[35]. Furthermore, further adjustments to training hyperparameters and more precise calibration of experimental signals will also yield improved imaging performance. These topics will serve as future research directions for this work.

Fig. 12 shows the qualitative images generated by TFM from the experimental measurement data of three specimens. The TFM images provide a moderate display of the defect where the wave first arrives, and its position is generally consistent with the inversion results of the proposed method. However, the overall geometry of the defect cannot be reconstructed. This is because TFM, based on ray approximation, is not fully utilizing the additional defect information provided by wave diffraction and scattering effects. It is noteworthy that the U-FNO-based ultrasonic FWI method can achieve quantitative reconstruction of the L-wave velocity, whereas the TFM method can only recover the signal amplitude in decibels.

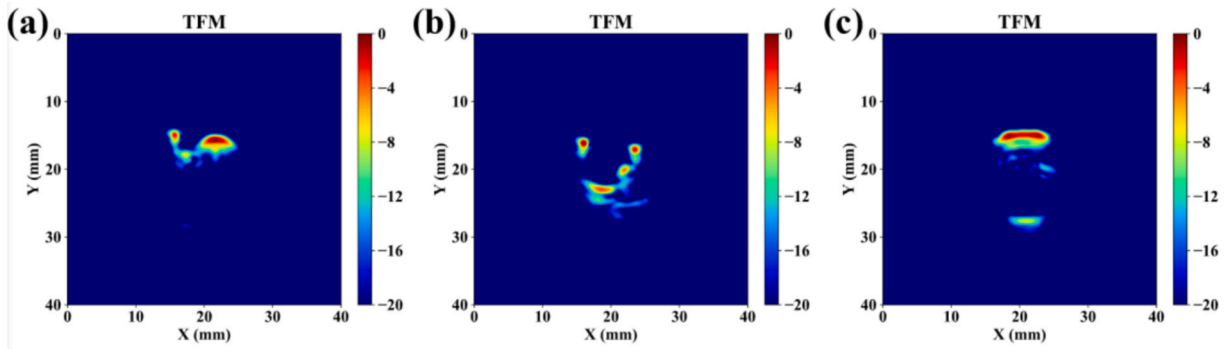


Fig. 12. Conventional TFM image. (a) TFM results corresponding to the model in Fig. 11(a). (b)TFM results corresponding to the model in Fig. 11 (d). (c) TFM results corresponding to the model in Fig.11(g).

4.3. Generalization ability

Generalization ability is a critical yet challenging requirement for data-driven full-waveform inversion methods. It provides an inverse operator for a sample set rather than a single inversion. In order to validate the network's capacity for generalization in training datasets constructed using the EMNIST letters, a digital geometric defect specimen was specifically designed, as illustrated in Fig. 11 (g), (h), and (i), which compare the inversion results of U-FNO and InversionNet, respectively. The findings demonstrate the efficacy of the proposed method in accurately predicting the internal structure of the sample and capturing defects, and possess a degree of generalization capability in velocity models that extends beyond the confines of the training datasets features. It is imperative to emphasize that the proposed method is not merely concerned with the reconstruction of defect geometry, but rather with elucidating the intricate nonlinear mapping relationship between FMC and the velocity distribution of the medium. The defect is characterized as a velocity anomaly region, and the entire specimen is treated as an inhomogeneous medium in data-driven FWI methods.

5. Discussion and limitations

This study demonstrates that the proposed U-FNO outperforms InversionNet across multiple metrics, including MAE, RMSE, SSIM, and L^2 relative error, on the publicly available EMNIST Letter dataset. While both methods achieve comparable performance on noise-free simulation data, InversionNet produces significantly inaccurate inversion results under noisy conditions (SNR = 6 dB). This limitation is further amplified when applied to real experimental data, where InversionNet fails to maintain robustness. In contrast, U-FNO consistently delivers superior inversion quality across both simulation and experimental scenarios. While the training of the U-FNO model entails considerable computational cost (~30 h), the trained model enables highly efficient inference, processing 1,000 samples in just 49 s, thereby providing a substantial speedup over conventional FWI methods and supporting near real-time imaging in FMC-based ultrasonic systems. The enhanced robustness and generalization of U-FNO can be attributed to its dual-path: The trainable kernel R, parameterized in Fourier space with truncated high-frequency modes, captures global frequency-domain features, while the Unet path restores localized high-frequency details. This synergistic design enables proficient handling of complex data patterns and subtle correlations. Furthermore, unlike traditional physics-based FWI, U-FNO eliminates the need for an initial velocity model and circumvents cycle-skipping issues. The present work addresses three major challenges in computational wave imaging: accuracy, efficiency and robustness, particularly in the context of ill-posed inverse problems with data available only on single-sided boundaries. Although the NDT problem serves as the main application scenario in this study, the proposed method is generalizable to other computational imaging domains that face similar inverse problems, such as medical ultrasound tomography, radar imaging, and seismic exploration.

Despite the considerable progress represented by the current numerical findings and laboratory experiments, a number of inherent limitations must be acknowledged.

First, the current modeling configuration is restricted to a velocity range of 1500–5800 m/s, with validation primarily conducted on water-filled defects embedded in a solid host medium. Although this represents a practically relevant inspection scenario, it does not encompass more extreme impedance contrasts, such as air-filled defects, where the higher velocity contrast significantly increases the difficulty of quantitative inversion. In such cases, the stronger scattering and mode-conversion effects necessitate more accurate modeling, which may require solving the elastic wave equations to account for mode conversion, employing finer spatial discretization to capture scattering in low-velocity regions, and additional inversion strategies to mitigate local minima and stabilize convergence, inevitably increasing computational cost and further complicating the inversion process. Second, the most critical challenge lies in the generalization capacity of the neural network under the supervised learning paradigm. Although the proposed method consistently achieves high-fidelity reconstruction within the training distribution, significant deviations in acquisition settings, material properties, or structural configurations may degrade performance. Such sensitivity to distribution shifts is inherent to supervised, data-driven inversion approaches. Therefore, while robust results are expected within the learned parameter space, extrapolation to substantially different scenarios remains challenging. To mitigate these limitations, several potential strategies have been proposed. First,

enhancing the diversity of the training dataset through mathematical randomization and geometric transformations, such as flipping, warping, and cropping of known velocity models may improve generalization. Second, transfer learning can be employed to accelerate adaptation to new domains by leveraging pre-trained knowledge. Third, incorporating physical constraints through an unsupervised framework guided by the governing wave equations may reduce data dependency[49,59], although this may compromise the benefits provided by learned inverse operators, including efficiency.

Currently, this work is limited to 2D inversion using ultrasound phased array techniques. However, the use of idealized 2D wave propagation models and point-source transducers restricts its applicability to three-dimensional (3D) imaging scenarios. Extending the approach to 3D requires more accurate modeling of wave physics, including the conversion between longitudinal and shear modes and the directionality of the transducers to reduce image distortion. Moreover, 3D imaging involves substantially larger observed datasets, which impose significant demands on computational resources. Advanced hardware and algorithmic optimizations will therefore be essential to support efficient processing and high-quality 3D reconstructions. Recent studies on FNO-based 3D wave propagation modeling have demonstrated the potential of neural operators to address high-dimensional inverse problems[60]. These insights will guide future work toward a neural operator-based framework for more efficient and scalable ultrasound imaging solutions across diverse applications.

6. Conclusion

In conclusion, we developed a U-FNO framework for direct inversion of ultrasonic full waveform data. By learning a nonlinear inverse mapping from FMC data to the L-wave velocity field, the framework removes the need for repeated forward wave simulations during each inversion and eliminates the reliance on having an initial velocity model and effectively circumvents cycle-skipping issues. At the inference stage, it is far more efficient than classical FWI. Benefiting from this efficient inference mechanism, the trained U-FNO model supports near real-time imaging in FMC-based ultrasonic systems. Unlike convolutional neural networks constrained by local receptive fields, U-FNO leverages Fourier-space kernel parameterization to capture global wave-physics features, thereby enabling high-fidelity reconstruction of complex geometric defects. Numerical simulations and experimental validations demonstrate that U-FNO consistently outperforms ray-based TFM and FCN-based InversionNet in terms of accuracy and robustness. Future work will focus on enhancing the generalization capability of the U-FNO framework, aiming to expand its applicability to a broader range of wave-based imaging scenarios.

CRediT authorship contribution statement

Jiang Lu: Writing – original draft, Validation, Software, Methodology, Investigation, Data curation, Conceptualization. **Lishuai Liu:** Writing – review & editing, Supervision, Funding acquisition, Formal analysis, Conceptualization. **Xuan Li:** Resources, Project administration, Formal analysis, Data curation. **Shijun Wang:** Software, Resources, Formal analysis. **Yanxun Xiang:** Writing – review & editing, Validation, Project administration, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

This work was supported by the National Natural Science Foundation of China (Grant Nos. 12422415, 12327807, 12374434, and 52321002) and the Shanghai Science and Technology Innovation Action Plan (No. 24TS1412200 and 24ZR1490900).

Data availability

Data will be made available on request.

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